

Kripa Sindhu

Software Engineer

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TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Go, Python, SQL

Frameworks & Libraries: React.js, Next.js, Angular, Node.js, Express.js, Tailwind CSS

Databases & Caching: PostgreSQL, MongoDB, Redis (sorted sets, Lua scripting), Firebase

DevOps & Observability: Docker, Docker Compose, GitHub Actions (CI/CD), AWS, Prometheus, Grafana, Linux

Tools & Concepts: Git, GitLab, REST APIs, Concurrency, Distributed Systems

WORK EXPERIENCE

BeatRoute Innovation

Software Engineer

May 2025 – Present

Gurugram, India

- Led the **migration of two production SaaS applications from Angular 14 to Angular 19**, modernizing the frontend architecture and cutting build times and bundle size with **zero production downtime** across **20k+ active users**.
- Architected the **Report Builder module** end-to-end — the company's flagship feature — building both the **Angular** configuration/visualization UI and the **Cube.js** query layer that powers dynamic report generation and export, driving a **30% increase** in self-serve report creation.
- Modeled and extended the **Cube.js semantic layer** over a multi-tenant **PostgreSQL** warehouse — authoring **cubes, measures, dimensions, and joins** — and designed **pre-aggregations** that reduced average query latency by **40%** on high-volume analytics workloads.
- Resolved **20+ high-impact data-layer edge cases** (missing per-tenant cubes, stale custom-field schemas, **row-level-security** gaps), hardening **multi-tenant data isolation** and cutting query-related support tickets by **30%**.
- Designed and implemented a **real-time chatbot interface** using **Angular** and **WebSockets**, enabling low-latency bi-directional communication (**sub-200ms** message delivery) for customers across **15+ countries**.
- Co-developed an **internal Angular UI component library** of **40+ reusable components** adopted across **5+ product teams**, reducing feature development effort by **30%** while enforcing consistent design and accessibility (**WCAG 2.1**) standards.
- Built a **unit and integration testing framework** with **Jasmine** and **Karma**, achieving **90%+ code coverage** across **5000+ test cases** and improving release confidence via automated CI pipelines.

PROJECTS

Distributed Task Queue System 🔄

Jan 2026 – Feb 2026

- Built a **distributed task processing platform** (**Go + Redis + Next.js 15** real-time dashboard) with priority scheduling, standalone worker nodes, and live node failure & recovery.
- Guaranteed **at-least-once delivery** under `kill -9` via lease-based visibility timeouts and **atomic Lua** ops, a reaper with **owner-fencing**, heartbeat failure detection (3s/10s), and **Redis-lease leader election** across replicas.
- Engineered a **concurrent worker pool** with **goroutines** and `sync.WaitGroup`, plus **graceful shutdown** via OS-signal trapping and context cancellation with a drain deadline for safe in-flight completion during deploys.
- Designed an **exponential backoff retry engine** with **dead-letter** isolation and **Redis sorted sets** ($O(\log N)$ dequeue); instrumented with **Prometheus/Grafana** and shipped via **multi-stage Docker**, Compose, and **GitHub Actions CI/CD**.

Feature Flag Management Platform 🔄

Feb 2026 – Mar 2026

- Architected a **full-stack feature flag platform** (**Go/Chi** backend, client SDK, **Next.js 16** dashboard) enabling dynamic feature rollouts across distributed services without redeployment.
- Implemented **real-time flag propagation** via **Server-Sent Events** + **Redis pub/sub**, delivering sub-second updates across horizontally scaled backend instances.
- Built a **deterministic percentage rollout engine** (**FNV-1a hashing**, $O(1)$ evaluation) consistent across Go and TypeScript, over a layered **Handler→Service→Repository** architecture with **PostgreSQL**, **Redis** caching, and API-key auth.

ACHIEVEMENTS

- Solved **1000+ Data Structures & Algorithms problems** across LeetCode, Codeforces, and GeeksforGeeks.
- Achieved **Knight** rank on **LeetCode** (contest rating 1850+, top ~5% globally).
- Qualified **GATE (Computer Science)** in both **2024 and 2025**, ranking in the **top 5%** of candidates nationwide.

PUBLICATION

- **EV-GREEN:** Co-authored a peer-reviewed journal article on a hybrid **MILP** + graph-based heuristic framework for electric-vehicle eco-routing with **GreenZone** prioritization and **Vehicle-to-Grid (V2G)** incentives; *Computing* (Springer Nature), Vol. 108, Art. 27, Feb 2026.

EDUCATION

Indian Institute of Information Technology, Guwahati

B.Tech in Computer Science and Engineering • CGPA: 7.39/10

Dec 2021 – Apr 2025

Assam, India